

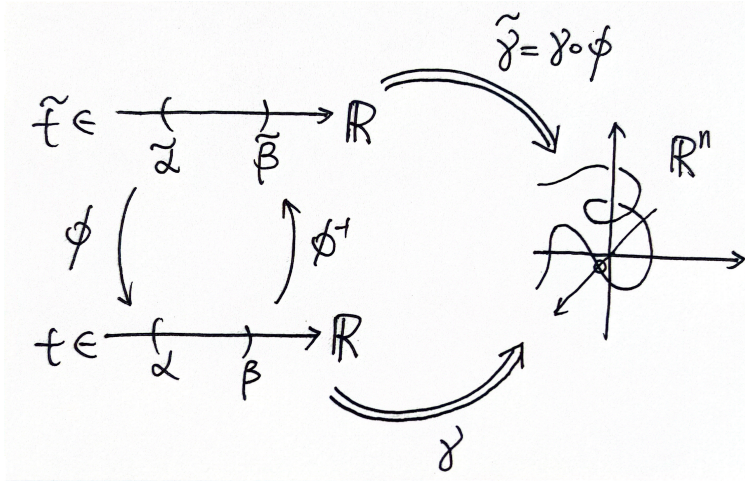
1 Reparametrization of a Curve

1.1 Reparametrization

Definition 1.(Reparametrization of a Curve)

For a parametrized curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$, its **reparametrization** is another parametrized curve $\tilde{\gamma} : (\tilde{\alpha}, \tilde{\beta}) \rightarrow \mathbb{R}^n$ with a diffeomorphism $\phi : (\tilde{\alpha}, \tilde{\beta}) \rightarrow (\alpha, \beta)$, i.e., both of ϕ and ϕ^{-1} are smooth, which satisfies

$$\tilde{\gamma}(\tilde{t}) = \gamma(\phi(\tilde{t})) \text{ for } \forall \tilde{t} \in (\tilde{\alpha}, \tilde{\beta}).$$



Remark: Since we see that

$$\gamma(t) = (\gamma \circ \phi)(\phi^{-1}(t)) = \tilde{\gamma}(\phi^{-1}(t)) \text{ for } \forall t \in (\alpha, \beta),$$

thus γ is also a reparametrization of $\tilde{\gamma}$.

Example 1. A parametrization of the circle $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ is $\gamma(t) = (\cos t, \sin t)$. By the diffeomorphism $\phi : t \mapsto \frac{\pi}{2} - t$, we get a reparametrization of the circle, which is

$$\tilde{\gamma}(t) \triangleq \gamma(\phi(t)) = \gamma\left(\frac{\pi}{2} - t\right) = \left(\cos\left(\frac{\pi}{2} - t\right), \sin\left(\frac{\pi}{2} - t\right)\right) = (\sin t, \cos t).$$

1.2 Regular Curves

Definition 2.(Regular Curves) A parametrized curve $\gamma(t)$ is called a **regular curve** if $\dot{\gamma}(t) \neq \mathbf{0}$ for $\forall t$. A point $\gamma(t_0)$ is called a **regular point** if $\dot{\gamma}(t_0) \neq \mathbf{0}$, otherwise it is called **singular point**.

Proposition 1. Let $\tilde{\gamma}(\tilde{t})$ be a reparametrization of the curve $\gamma(t)$, if $\gamma(t)$ is regular, then $\tilde{\gamma}(\tilde{t})$ is also regular.

Proof: Since $\tilde{t} = \phi^{-1}(t)$ and $\phi(\phi^{-1}(t)) = t$, hence by the chain rule we get

$$\frac{d\phi}{d\tilde{t}} \cdot \frac{d(\phi^{-1})}{dt} = 1,$$

which implies that $\frac{d\phi}{d\tilde{t}} \neq 0$ for $\forall t$. so take the derivative of $\tilde{\gamma}(\tilde{t}) = \gamma(\phi(\tilde{t}))$ w.r.t. $\tilde{t}$, we get

$$\frac{d\tilde{\gamma}}{d\tilde{t}} = \frac{d\gamma}{dt} \cdot \frac{d\phi}{d\tilde{t}} \neq 0,$$

which means $\tilde{\gamma}(\tilde{t})$ is also regular. ■

Recall that for a parametrized curve $\gamma(t)$, its arc-length is given by $s(t) = \int_{t_0}^t \|\dot{\gamma}(u)\| du$. It can be easily shown that $s(t)$ is a smooth function of t . The important result is the following.

Theorem 1. A parametrized curve γ has a **unit-speed reparametrization** $\iff \gamma$ is regular.

Proof: ($\implies$) Suppose $\tilde{\gamma}(\tilde{t})$ is a unit-speed reparametrization of $\gamma(t)$ whose reparametrization map is $\phi : (\tilde{\alpha}, \tilde{\beta}) \longrightarrow (\alpha, \beta)$, $\tilde{t} \mapsto t = \phi(\tilde{t})$, i.e., $\tilde{\gamma}(\tilde{t}) = \gamma(\phi(\tilde{t})) = \gamma(t)$. So we get

$$\left\| \frac{d\tilde{\gamma}}{d\tilde{t}} \right\| = \left\| \frac{d\gamma}{dt} \cdot \frac{d\phi}{d\tilde{t}} \right\| = \left| \frac{d\phi}{d\tilde{t}} \right| \cdot \left\| \frac{d\gamma}{dt} \right\| = 1,$$

which implies that

$$\left\| \frac{d\gamma}{dt} \right\| \neq 0 \text{ for } \forall t,$$

hence $\gamma(t)$ is regular;

($\impliedby$) Suppose that $\left\| \frac{d\gamma}{dt} \right\| \neq 0$ for $\forall t$. Let $s(t)$ be the arc-length function of $\gamma(t)$, i.e.,

$$s(t) = \int_{\alpha}^t \|\dot{\gamma}(u)\| du \text{ for } \forall t \in (\alpha, \beta),$$

so we have

$$\frac{ds}{dt} = \|\dot{\gamma}(t)\| \neq 0,$$

note that $s(t)$ is smooth, so by the inverse function theorem in one-variable case, we know that $s : (\alpha, \beta) \longrightarrow s((\alpha, \beta)) \subseteq \mathbb{R}$ is a diffeomorphism. we denote $(\tilde{\alpha}, \tilde{\beta}) \triangleq s((\alpha, \beta))$, now we take the reparametrization map as $\phi := s^{-1} : (\tilde{\alpha}, \tilde{\beta}) \longrightarrow (\alpha, \beta)$, so we get

$$\tilde{\gamma}(s) = (\gamma \circ \phi)(s) = (\gamma \circ s^{-1})(s(t)) = \gamma(t),$$

thus

$$\frac{d\gamma}{dt} = \frac{d\tilde{\gamma}}{ds} \cdot \frac{ds}{dt},$$

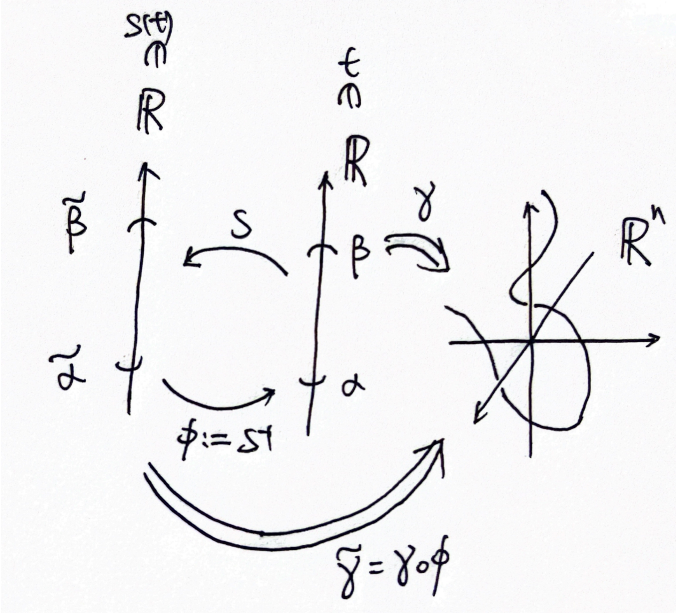
note that $\frac{ds}{dt} = \|\dot{\gamma}\|$, hence

$$\left\| \frac{d\gamma}{dt} \right\| = \left\| \frac{d\tilde{\gamma}}{ds} \cdot \frac{ds}{dt} \right\| = \left\| \frac{d\tilde{\gamma}}{ds} \right\| \cdot \left\| \frac{ds}{dt} \right\|,$$

i.e., we see that

$$\left\| \frac{d\tilde{\gamma}}{ds} \right\| \equiv 1,$$

which completes the proof. ■



Remark: From this theorem we see that for a regular curve $\gamma(t)$, the arc-length is essentially the only unit-speed parameter for it.

Corollary 1. Let γ be a regular curve and let $\tilde{\gamma}$ be a unit-reparametrization of γ such that

$$\tilde{\gamma}(u(t)) \equiv \gamma(t) \text{ and } \left\| \frac{d\tilde{\gamma}}{du} \right\| \equiv 1,$$

where $u(t)$ is smooth function of t . If $s(t)$ is the arc-length function of $\gamma(t)$ starting at some point, then

$$u(t) = s(t) \pm c,$$

where c is a constant. Conversely, if $u(t) = \pm s(t) + c$, then $\tilde{\gamma}(u)$ is a unit-speed reparametrization of $\gamma(t)$.

Proof: Just note that

$$\left\| \frac{d\gamma}{dt} \right\| = \left\| \frac{d\tilde{\gamma}}{du} \cdot \frac{du}{dt} \right\|,$$

thus $\left\| \frac{d\tilde{\gamma}}{du} \right\| \equiv 1$ iff

$$\left| \frac{du}{dt} \right| = \left\| \frac{d\gamma}{dt} \right\|,$$

i.e.,

$$\frac{du}{dt} = \pm \left\| \frac{d\gamma}{dt} \right\|,$$

hence we get

$$u(t) = \int \frac{du}{dt} dt = \int \pm \left\| \frac{d\gamma}{dt} \right\| dt = \pm s(t) + c.$$

which completes the proof. ■

Example 2.(unit-speed reparametrization of logarithmic spiral)

Consider the logarithmic spiral

$$\gamma(t) := (e^{kt} \cos t, e^{kt} \sin t),$$

so we get

$$\|\dot{\gamma}(t)\|^2 = (k^2 + 1)e^{2kt} > 0,$$

so $\gamma(t)$ is a regular curve. Let the starting point of $\gamma(t)$ be $\gamma(0) = (1, 0)$, then the arc-length $s(t)$ is given by

$$s(t) = \int_0^t \|\dot{\gamma}(u)\| du = \frac{\sqrt{k^2 + 1}}{k}(e^{kt} - 1).$$

Thus we get

$$t = \frac{1}{k} \ln \left(\frac{ks}{\sqrt{k^2 + 1}} + 1 \right),$$

so the unit-speed reparametrization of γ is given by

$$\begin{aligned} \tilde{\gamma}(s) = & \left(\left(\frac{ks}{\sqrt{k^2 + 1}} + 1 \right) \cos \left(\frac{1}{k} \ln \left(\frac{ks}{\sqrt{k^2 + 1}} + 1 \right) \right), \left(\frac{ks}{\sqrt{k^2 + 1}} + 1 \right) \sin \left(\frac{1}{k} \ln \left(\frac{ks}{\sqrt{k^2 + 1}} + 1 \right) \right) \right). \end{aligned}$$

Example 3.(unit-speed reparametrization that can't be written down explicitly)

The *twisted cubic*, which is

$$\gamma(t) := (t, t^2, t^3), \quad t \in \mathbb{R}.$$

So we get

$$\dot{\gamma}(t) = (1, 2t, 3t^2), \quad \|\dot{\gamma}(t)\| = \sqrt{1 + 4t^2 + 9t^4} \neq 0 \text{ for } \forall t \in \mathbb{R},$$

so $\gamma(t)$ is regular, so it can be unit-speed reparametrized, but the arc-length $s(t)$ is

$$s(t) = \int_0^t \sqrt{1 + 4u^2 + 9u^4} du,$$

which cannot be evaluated by elementary functions.